Quiz 7

Friday, October 23 — online in Canvas — transforming space (Lecture 20)

Last updated: October 23, 2026.

This quiz is online in Canvas on Friday, October 23. Practice from this page, and submit the quiz in Canvas.

Textbook: Boyd & Vandenberghe, *VMLS* (PDF)

Covers the week before only: Transforms (taught Wed Oct 14 and Fri Oct 16). Same-week Lectures 21–23 and Lab 2 are **not** on this quiz.

Textbooks: Boyd & Vandenberghe (VMLS) · Lachniet Octave PDF · also on Resources.

Schedule

Draw on paper for every item that says draw. Label the original and the image, and mark e_1 and e_2 when you use them. A point lands at the tip of A times that vector. The columns of A are where the standard basis lands: column j is Ae_j.

Core

Fluency (complete prompts — practice these by hand)

C7.1. Let

$$A = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}.$$

Compute Ae_1 and Ae_2 for $e_1 = (1, 0)$ and $e_2 = (0, 1)$. In one sentence, how do those two images sit inside A?

C7.2. Reflection through the x_1-axis sends $e_1 = (1, 0) \mapsto (1, 0)$ and $e_2 = (0, 1) \mapsto (0, -1)$. Write A, compute $A(2, 3)$, and draw e_1, e_2, and both images on one set of axes.

C7.3. Rotation by 90° counterclockwise. Write the matrix. Where do e_1 and e_2 land? Draw those two arrows, then draw the unit square with vertices $(0, 0)$, $(1, 0)$, $(1, 1)$, $(0, 1)$ and its image.

C7.4. Stretch by $A = \text{diag}(2, 1/2)$. Where do e_1 and e_2 land? Draw the same unit square and its image.

C7.5. Slant (shear) by

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}.$$

Where do e_1 and e_2 land? Draw the unit square and its image. What shape is the image?

C7.6. Flip through the x_2-axis: build A by tracking e_1 and e_2. The triangle has vertices $(0, 0)$, $(2, 0)$, and $(1, 1)$. Compute the three image points and draw the original triangle and the image.

C7.7. Let $X = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$ (columns are three vertices) and $A = \text{diag}(2, 1)$. Compute AX. Draw the original triangle and the image, and describe the effect in one sentence.

C7.8. The rectangle has vertices $(0, 0)$, $(2, 0)$, $(2, 1)$, $(0, 1)$. Apply the 90° counterclockwise rotation from C7.3 to each vertex. Draw the original rectangle and the image.

C7.9. Using the shear in C7.5, where does the single point $(0, 2)$ land? You may use that $(0, 2) = 2e_2$. Draw the point and its image.

Textbook Core

- B 7.2, 7.3, 7.4

Stretch

- B 7.1, 7.8
- Sketch the unit circle $x_1^2 + x_2^2 = 1$ and its image under $A = \mathrm{diag}(2, 1/2)$. You do not need a perfect curve: mark where e_1, $-e_1$, e_2, and $-e_2$ go, and join them.

What a quiz item might change

Which map (rotation, flip, stretch, or slant), which shape (basis arrows, square, triangle, or rectangle), or asking only for the images of e_1 and e_2 before any other points.